TECHNICAL SPECIFICATIONS

NUCLEAR DATA INC
100 West Golf Road
Palatine, Illinois 60067
Phone A.C. 312 • 529-4600



SERIES 3300 ANALOG TO DIGITAL CONVERTER

The Analog to Digital Converter Module (ADC) is designed primarily for processing of amplitude-modulated signal pulses, such as encountered in nuclear pulse height analysis. It contains an analog-to-digital converter and a twelve-bit address scaler that provide resolution of up to 4096 channels in-line.

Data acquisition efficiency is enhanced by a 16 megacycle digitizing rate which significantly reduces ADC dead time effects. The digitizing oscillator is crystal-controlled to ensure high long term stability. If desired, the ADC can be operated to accept input signals only when externally generated coincidence gating pulses are supplied.

Analog Conversion Gains

128, 256, 512, 1024, 2048, and 4096 addresses full scale.

Direct Access Signal Input

Pulse Requirements

- Amplitude: 0 to 10 volts nominal at minimum zero level.
- Polarity: Positive.
- Rise Time: 0.1 microsecond minimum, 5 microseconds maximum.
- Duration: 0.5 to 3 microseconds.

Slow A.C. and D.C. Signal Requirements

- Amplitude: 0 to 10 volts nominal at minimum zero level.
- Polarity: Positive (a D.C. offset can be introduced for analysis of bipolar signals of up to 10 volts peak-to-peak amplitude).

Input Impedance — Approximately 2 kilohms.

10 Volt Signal Input (Delayed)

Pulse Requirements

- Amplitude: 0 to 10 volts nominal at minimum zero level.
- Polarity: Positive or bipolar.
- Rise Time: 0.1 microsecond minimum, 5 microseconds maximum, exponential.
- Duration: 0.5 to 3 microseconds.

Internal Delay — 1.25 microseconds.
Input Impedance — Approximately 2.2 kilohms.

100 Volt Signal Input (Delayed)

Pulse Requirements

- Amplitude: 0 to 100 volts nominal at minimum zero level.
- Polarity: Positive or bipolar.
- Rise Time: 0.1 microsecond minimum, 5 microseconds maximum, exponential.

Duration: 0.5 to 3 microseconds.
Internal Delay — 1.25 microseconds.
Input Impedance — Approximately 22 kilohms.

Coincidence Input

Signal Requirements
Amplitude: +2 to +8 volts.
Polarity: Positive.
Duration: 0.1 microsecond minimum.
Input Impedance — Approximately 1 milliampere SINK (DTL Logic).

Conversion Time (PHA)

.0625N microsecond, where N is equal to the number of address advances produced for a given amplitude or duration, plus X ÷ 512 microseconds, where X is equal to the selected analog conversion gain.

Linearity

Integral
Better than 0.1% over the top 99% of full scale.

Differential
Less than 1% deviation from mean channel width over top 99% of full scale.

Zero Level

Adjustable from 0 to 100% of full scale.

Threshold

Adjustable from 0 to 25% of full scale, as selected by a front panel control.

Upper Level

Adjustable from 75% to 100% of full scale, as selected by a front panel control.

Stability

Time
Less than one channel drift per day at stable ambient temperature.
Temperature
Less than 0.1% zero drift and 0.2% gain drift per 5°C increase from 15°C to 40°C.